

Early Forest Fire Detection Using Satellite Imagery

Sparsh Kumar | B.Tech CSE | AIML Project

Abstract

Forest fires pose a serious environmental threat causing ecological destruction, economic loss, and human casualties. Early detection is critical to reduce fire spread and improve disaster response efficiency. This research investigates deep learning-based approaches for early wildfire detection using satellite imagery from MODIS and VIIRS sensors. We analyze convolutional neural networks (CNN), CNN-LSTM hybrid architectures, and Vision Transformers (ViT/Swin) for spatial-temporal fire pattern recognition. The study identifies research gaps in multi-sensor fusion and early ignition-stage detection and proposes a hybrid transformer-temporal framework for improved near-real-time wildfire detection.

1. Introduction

Satellite-based wildfire monitoring systems such as MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite) provide continuous global observations of thermal anomalies. Traditional threshold-based fire detection methods often suffer from high false positives due to atmospheric interference, sun-glint, and industrial heat sources. Recent advancements in Artificial Intelligence and Deep Learning enable automated feature extraction from multispectral imagery, improving detection accuracy.

2. Literature Review

Recent studies demonstrate that Convolutional Neural Networks (CNNs) effectively extract spatial fire signatures from multispectral bands. However, CNNs lack inherent temporal modeling. To address this limitation, researchers proposed CNN-LSTM hybrid models where CNN extracts spatial features and LSTM models fire spread over time. More recently, Vision Transformers (ViT) and Swin Transformers have shown superior performance due to their ability to capture long-range spatial dependencies and global contextual relationships.

3. Dataset Description

The MODIS Active Fire Product provides global thermal anomaly detection at moderate spatial resolution, while VIIRS offers improved spatial resolution (375m) for fire hotspot detection. These

datasets include fire confidence levels, brightness temperature, and geolocation data. For model training, satellite scenes are converted into image patches with corresponding fire labels.

4. Proposed Methodology

This project proposes a hybrid architecture integrating multi-sensor fusion and temporal modeling. A Vision Transformer backbone extracts spatial features from stacked MODIS and VIIRS bands. Temporal sequences are processed using LSTM layers to capture fire spread dynamics. Class imbalance is addressed using focal loss, and evaluation metrics include precision, recall, F1-score, IoU, and detection latency.

5. Research Gap

Despite significant progress, limited research focuses specifically on early ignition-stage detection. Multi-sensor fusion techniques remain underexplored, and transformer-based temporal architectures are still in early development stages. Reducing detection latency while maintaining high precision remains a key challenge.

6. Conclusion

Deep learning significantly enhances satellite-based wildfire detection systems. By combining transformer-based spatial modeling with temporal sequence learning and multi-sensor fusion, future systems can achieve improved early-stage detection accuracy and real-time deployment capability. This research lays the foundation for developing an advanced wildfire monitoring framework suitable for operational disaster management systems.

7. References

[1] Saleh et al., "Forest fire surveillance systems: A review of deep learning methods," 2023. [2] Kang et al., "Deep learning model using satellite data to reduce wildfire detection latency," 2022. [3] Recent Vision Transformer-based wildfire detection studies (2023–2025).